

Disproof of Conjecture 00000001028: there are not 21 exceptional values below 316 for Kirkman quadruple systems

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2026-10-04

Abstract

Conjecture 00000001028 asserts that Kirkman quadruple systems $\text{KQS}(v)$ (resolvable $S(2, 4, v)$ designs) exist for all $v \equiv 4 \pmod{12}$ with $v \geq v_0 = 316$, and that below v_0 there are exactly 21 exceptional values. Only 26 values $v \equiv 4 \pmod{12}$ lie below 316. We construct explicit Kirkman quadruple systems for the seven values $v = 4, 16, 28, 40, 52, 64, 76$. Hence at most 19 values below 316 are exceptional (at most 20 even if $v_0 = 316$ itself is counted), and the second clause of the conjecture is false. The seven designs, the bound and the negation of the conjecture are formalized and machine-checked in Lean 4 (core library only). (In fact, by a 1972 theorem of Hanani, Ray-Chaudhuri and Wilson, there are no exceptional values at all.)

1 The conjecture and its reading

Definition: A resolvable $S(2, 4, v)$ (Kirkman quadruple system) is a design whose blocks partition into parallel classes. *Conjecture:* There is an explicit threshold $v_0 = 316$ such that $\text{KQS}(2, 4, v)$ exist for all $v \equiv 4 \pmod{12}$ with $v \geq v_0$; below v_0 there are exactly 21 exceptional values.

The Chinese version says the same. The conjecture is a conjunction of

(C1) $\text{KQS}(v)$ exists for every $v \equiv 4 \pmod{12}$ with $v \geq 316$, and

(C2) below $v_0 = 316$ there are exactly 21 exceptional values.

We refute (C2); this refutes the conjecture regardless of (C1).

Exceptional values. In the context of (C1), an *exceptional value* is an admissible order $v \equiv 4 \pmod{12}$, $v < v_0$, for which no $\text{KQS}(v)$ exists. (C2) says that the set of these values has exactly 21 elements. We show that it has at most 19 elements. We also cover the following variants.

- *Non-strict “below”* ($v \leq 316$): at most 20 exceptional values, so still not 21. (Under (C1), $v = 316$ is not exceptional anyway.)
- *Excluding trivial orders* (for example, ignoring $v = 4$): the set only gets smaller, so the bound ≤ 19 still holds.
- *Any residue:* if “exceptional” meant any $v < 316$ without a KQS, regardless of the residue of v , then the count would be at least 288 (Remark 2), again not 21. This variant is discussed only in this report; the Lean development treats the main reading and the non-strict variant.

- “*Exceptional*” as “*existence undecided*”: this is not a property of v alone. In any case, the seven values below are settled by explicit, verified designs, so at most 19 values could be undecided. By [1], none is.

2 Definitions

Definition 1. Let $v \geq 1$ and $V = \{0, 1, \dots, v-1\}$. A *Kirkman quadruple system* $\text{KQS}(v)$ is a family $\mathcal{R} = (P_1, \dots, P_r)$ of *parallel classes*. Each P_i is a family of *blocks*, and the following hold:

1. every block is a 4-element subset of V ;
2. every parallel class partitions V : each point of V lies in exactly one block of P_i ;
3. every pair of distinct points of V lies in exactly one block of $P_1 \cup \dots \cup P_r$.

Conditions 1 and 3 say that the blocks form an $S(2, 4, v)$, i.e. a $2-(v, 4, 1)$ design; condition 2 says that $\mathcal{R}$ is a resolution of it.

Remark 2. If a $\text{KQS}(v)$ exists and $v \geq 2$, then $v \equiv 4 \pmod{12}$. Indeed, the blocks through a fixed point x partition $V \setminus \{x\}$ into triples, so $3 \mid v-1$. A parallel class partitions V into 4-sets, so $4 \mid v$. Hence there are exactly 26 admissible values below 316, namely 4, 16, 28, $\dots$, 304, and 27 up to 316. Moreover, every $v < 316$ with $2 \leq v$ and $v \not\equiv 4 \pmod{12}$ has no KQS . There are $314 - 26 = 288$ such values, which is why the “any residue” reading gives at least 288 exceptions.

3 Seven Kirkman quadruple systems

3.1 $v = 4$

One parallel class consisting of the single block $\{0, 1, 2, 3\}$.

3.2 $v = 16$ and $v = 64$: affine spaces over $\text{GF}(4)$

Let $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$, encoded as $\{0, 1, 2, 3\}$; addition is bitwise XOR and $\omega \cdot \omega = \omega^2$. The points are the vectors of $\mathbb{F}_4^n$ ($n = 2, 3$), and $(a_0, \dots, a_{n-1})$ is encoded as $\sum a_i 4^i$. The blocks are the affine lines $\{p + td : t \in \mathbb{F}_4\}$ with $d \neq 0$. Two distinct points p, q lie on exactly one line, namely the one with direction $q - p$, since the direction of a line through p and q is a nonzero multiple of $q - p$. For each direction d (up to scalars there are $(4^n - 1)/3$ of them), the lines with direction d are the cosets of $\mathbb{F}_4 d$ and partition the space. So the lines of $\text{AG}(n, 4)$ form a $\text{KQS}(4^n)$ with $(4^n - 1)/3$ parallel classes: 5 classes of 4 blocks for $v = 16$ and 21 classes of 16 blocks for $v = 64$.

3.3 $v = 28$: the Hermitian unital

Let $\mathbb{F}_9 = \mathbb{F}_3[i]$ with $i^2 = -1$. The Hermitian curve $x^4 + y^4 + z^4 = 0$ in $\text{PG}(2, 9)$ has 28 points. Every line of $\text{PG}(2, 9)$ meets it in 1 or 4 points, so the 63 secant lines form a $2-(28, 4, 1)$ design, the Hermitian unital $U(3)$. An exact-cover search found a resolution of it into 9 parallel classes of 7 blocks. The labelling is as follows. We write $a + bi$ as the pair $(a, b) \in \{0, 1, 2\}^2$, normalize projective points so that the first nonzero coordinate is 1, and number the 28 unital points $0, \dots, 27$ in lexicographic order of their coordinate triples. The parallel classes are:

P_1 : $\{0, 13, 16, 26\} \{1, 5, 11, 12\} \{2, 4, 17, 25\} \{3, 9, 21, 22\} \{6, 8, 19, 20\} \{7, 15, 23, 27\} \{10, 14, 18, 24\}$
 P_2 : $\{0, 7, 10, 12\} \{1, 13, 20, 22\} \{2, 9, 16, 27\} \{3, 6, 18, 23\} \{4, 8, 24, 26\} \{5, 14, 17, 21\} \{11, 15, 19, 25\}$
 P_3 : $\{0, 5, 19, 22\} \{1, 8, 17, 27\} \{2, 6, 10, 13\} \{3, 15, 20, 24\} \{4, 12, 18, 21\} \{7, 9, 25, 26\} \{11, 14, 16, 23\}$
 P_4 : $\{0, 1, 2, 3\} \{4, 14, 22, 27\} \{5, 10, 23, 26\} \{6, 15, 16, 21\} \{7, 11, 17, 20\} \{8, 13, 18, 25\} \{9, 12, 19, 24\}$
 P_5 : $\{0, 8, 21, 23\} \{1, 7, 16, 24\} \{2, 14, 19, 26\} \{3, 4, 11, 13\} \{5, 9, 18, 20\} \{6, 12, 25, 27\} \{10, 15, 17, 22\}$
 P_6 : $\{0, 6, 17, 24\} \{1, 4, 19, 23\} \{2, 7, 18, 22\} \{3, 5, 16, 25\} \{8, 9, 10, 11\} \{12, 13, 14, 15\} \{20, 21, 26, 27\}$
 P_7 : $\{0, 11, 18, 27\} \{1, 6, 9, 14\} \{2, 5, 8, 15\} \{3, 12, 17, 26\} \{4, 10, 16, 20\} \{7, 13, 19, 21\} \{22, 23, 24, 25\}$
 P_8 : $\{0, 4, 9, 15\} \{1, 10, 21, 25\} \{2, 12, 20, 23\} \{3, 7, 8, 14\} \{5, 13, 24, 27\} \{6, 11, 22, 26\} \{16, 17, 18, 19\}$
 P_9 : $\{0, 14, 20, 25\} \{1, 15, 18, 26\} \{2, 11, 21, 24\} \{3, 10, 19, 27\} \{4, 5, 6, 7\} \{8, 12, 16, 22\} \{9, 13, 17, 23\}$

3.4 $v = 40$: a packing of PG(3,3)

The 40 points and 130 lines of PG(3, 3) form a 2-(40, 4, 1) design (two points span one line). A *spread* is a set of 10 pairwise disjoint lines; a *packing* is a partition of the 130 lines into 13 spreads, i.e. a resolution. Packings of PG(3, q) exist for all q (Denniston [2]). An exact-cover search gave the following one. Points are the vectors of $\{0, 1, 2\}^4 \setminus \{0\}$ with first nonzero coordinate 1, numbered $0, \dots, 39$ in lexicographic order.

P_1 : $\{0, 34, 35, 36\} \{1, 6, 9, 12\} \{2, 15, 16, 20\} \{3, 22, 27, 29\} \{4, 21, 30, 39\}$
 $\{5, 13, 23, 33\} \{7, 14, 26, 38\} \{8, 18, 28, 32\} \{10, 19, 25, 31\} \{11, 17, 24, 37\}$
 P_2 : $\{0, 25, 26, 27\} \{1, 5, 8, 11\} \{2, 32, 36, 37\} \{3, 15, 17, 19\} \{4, 13, 22, 31\}$
 $\{6, 20, 28, 39\} \{7, 18, 30, 33\} \{9, 21, 23, 34\} \{10, 14, 29, 35\} \{12, 16, 24, 38\}$
 P_3 : $\{0, 31, 32, 33\} \{1, 22, 25, 28\} \{2, 13, 17, 21\} \{3, 5, 7, 12\} \{4, 20, 29, 38\}$
 $\{6, 18, 26, 34\} \{8, 14, 27, 37\} \{9, 19, 24, 35\} \{10, 15, 30, 36\} \{11, 16, 23, 39\}$
 P_4 : $\{0, 19, 20, 21\} \{1, 32, 35, 38\} \{2, 22, 26, 30\} \{3, 4, 9, 11\} \{5, 14, 24, 31\}$
 $\{6, 17, 25, 36\} \{7, 15, 27, 39\} \{8, 16, 29, 33\} \{10, 13, 28, 34\} \{12, 18, 23, 37\}$
 P_5 : $\{0, 7, 8, 9\} \{1, 31, 34, 37\} \{2, 23, 27, 28\} \{3, 14, 16, 21\} \{4, 17, 26, 35\}$
 $\{5, 15, 22, 32\} \{6, 19, 30, 38\} \{10, 18, 24, 39\} \{11, 13, 29, 36\} \{12, 20, 25, 33\}$
 P_6 : $\{0, 37, 38, 39\} \{1, 13, 16, 19\} \{2, 6, 7, 11\} \{3, 23, 25, 30\} \{4, 15, 24, 33\}$
 $\{5, 17, 27, 34\} \{8, 21, 22, 35\} \{9, 18, 29, 31\} \{10, 20, 26, 32\} \{12, 14, 28, 36\}$
 P_7 : $\{0, 1, 2, 3\} \{4, 18, 27, 36\} \{5, 19, 29, 39\} \{6, 13, 24, 32\} \{7, 20, 23, 35\}$
 $\{8, 15, 25, 38\} \{9, 17, 28, 33\} \{10, 16, 22, 37\} \{11, 14, 30, 34\} \{12, 21, 26, 31\}$
 P_8 : $\{0, 10, 11, 12\} \{1, 33, 36, 39\} \{2, 24, 25, 29\} \{3, 13, 18, 20\} \{4, 14, 23, 32\}$
 $\{5, 21, 28, 38\} \{6, 16, 27, 35\} \{7, 19, 22, 34\} \{8, 17, 30, 31\} \{9, 15, 26, 37\}$
 P_9 : $\{0, 13, 14, 15\} \{1, 24, 27, 30\} \{2, 4, 8, 12\} \{3, 32, 34, 39\} \{5, 18, 25, 35\}$
 $\{6, 21, 29, 37\} \{7, 16, 28, 31\} \{9, 20, 22, 36\} \{10, 17, 23, 38\} \{11, 19, 26, 33\}$
 P_{10} : $\{0, 22, 23, 24\} \{1, 15, 18, 21\} \{2, 33, 34, 38\} \{3, 6, 8, 10\} \{4, 19, 28, 37\}$
 $\{5, 16, 26, 36\} \{7, 17, 29, 32\} \{9, 14, 25, 39\} \{11, 20, 27, 31\} \{12, 13, 30, 35\}$
 P_{11} : $\{0, 4, 5, 6\} \{1, 23, 26, 29\} \{2, 14, 18, 19\} \{3, 31, 36, 38\} \{7, 13, 25, 37\}$
 $\{8, 20, 24, 34\} \{9, 16, 30, 32\} \{10, 21, 27, 33\} \{11, 15, 28, 35\} \{12, 17, 22, 39\}$
 P_{12} : $\{0, 28, 29, 30\} \{1, 14, 17, 20\} \{2, 5, 9, 10\} \{3, 33, 35, 37\} \{4, 16, 25, 34\}$
 $\{6, 15, 23, 31\} \{7, 21, 24, 36\} \{8, 13, 26, 39\} \{11, 18, 22, 38\} \{12, 19, 27, 32\}$
 P_{13} : $\{0, 16, 17, 18\} \{1, 4, 7, 10\} \{2, 31, 35, 39\} \{3, 24, 26, 28\} \{5, 20, 30, 37\}$
 $\{6, 14, 22, 33\} \{8, 19, 23, 36\} \{9, 13, 27, 38\} \{11, 21, 25, 32\} \{12, 15, 29, 34\}$

3.5 $v = 52$ and $v = 76$: 1-rotational designs

Lemma 3. Let $m \geq 2$, $N = 3m$, and let $B_1, \dots, B_s \subset \mathbb{Z}_N$ be 4-sets ($4s = m - 1$) such that

(a) the $4s$ elements of $B_1 \cup \dots \cup B_s$, reduced mod m , are exactly $1, 2, \dots, m - 1$, each once;

(b) the 12s differences $x - y$ ($x \neq y$ in the same B_i) are exactly the elements of $\mathbb{Z}_N \setminus \{0, m, 2m\}$, each once.

On $V = \mathbb{Z}_N \cup \{\infty\}$, for $j = 0, \dots, m-1$ let $P_j = \{\{j, j+m, j+2m, \infty\}\} \cup \{B_i + j + mk : 1 \leq i \leq s, 0 \leq k \leq 2\}$. Then $(P_0, \dots, P_{m-1})$ is a KQS($3m+1$).

Proof. Classes. By (a), the sets $B_i + mk$ ($k = 0, 1, 2$) cover each residue class $r \not\equiv 0 \pmod{m}$ of $\mathbb{Z}_N$ exactly once (the element $x \equiv r$ yields $x, x+m, x+2m$). After shifting by j , they cover every residue class except that of j , and $\{j, j+m, j+2m, \infty\}$ covers the rest. *Pairs.* The pair $\{\infty, y\}$ lies only in $\{j, j+m, j+2m, \infty\}$ with $j \equiv y \pmod{m}$. Let $y \neq z$ be finite and $d = z - y$. If $d \in \{m, 2m\}$, then $y \equiv z \pmod{m}$; by (a), no translate of a B_i contains two elements of the same residue class mod m , so the only block containing y, z is the ∞ -block of $j \equiv y$. Otherwise, by (b), $d = x - x'$ for a unique ordered pair (x', x) in a unique B_i . A translate $B_i + t$ contains y, z exactly when $t = y - x'$, and no other $B_{i'} + t'$ contains them, by uniqueness in (b). Finally, the translates $B_i + j + mk$ ($0 \leq j < m, 0 \leq k < 3$) are the N translates $B_i + t, t \in \mathbb{Z}_N$, each once. Thus every pair lies in exactly one block. $\square$

A backtracking search gave base blocks satisfying (a) and (b).

- $m = 17, v = 52$ ($\infty = 51$): $\{1, 2, 20, 24\}, \{4, 25, 12, 49\}, \{5, 10, 45, 30\}, \{6, 9, 48, 50\}$.
- $m = 25, v = 76$ ($\infty = 75$): $\{1, 34, 47, 48\}, \{2, 33, 67, 69\}, \{3, 56, 7, 71\}, \{4, 61, 13, 16\}, \{5, 60, 37, 43\}, \{14, 65, 70, 49\}$.

For instance, for $m = 17$ the residues mod 17 of the four base blocks are

$$\{1, 2, 3, 7\}, \{4, 8, 12, 15\}, \{5, 10, 11, 13\}, \{6, 9, 14, 16\}.$$

Conditions (a) and (b) are checked by `verify.py`. Independently, the Lean development checks the resulting designs directly (17 classes and 221 blocks for $v = 52$; 25 classes and 475 blocks for $v = 76$).

4 The disproof

Theorem 4. *There are at most 19 values $v \equiv 4 \pmod{12}$, $v < 316$, without a KQS(v), and at most 20 with $v \leq 316$. Hence clause (C2) and Conjecture 00000001028 are false.*

Proof. The admissible values below 316 are $4 + 12k$, $0 \leq k \leq 25$, i.e. 26 values. By Section 3, KQS(v) exists for $v \in \{4, 16, 28, 40, 52, 64, 76\}$. So every exceptional value lies in the remaining set $\{88, 100, \dots, 304\}$ of 19 values (together with 316 in the non-strict reading: 20 values). A set of exactly 21 exceptional values would have to be contained in a set with at most 20 elements, which is impossible. $\square$

Remark 5. The truth is much stronger. Hanani, Ray-Chaudhuri and Wilson [1] proved that a KQS(v) exists for *every* $v \equiv 4 \pmod{12}$, so there are no exceptional values at all. Our disproof does not depend on this theorem. It uses only the seven explicit, machine-verified designs above.

5 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project (core library only, no Mathlib). The main file is `Main.lean`.

- **Resolution** is a list of parallel classes; a class is a list of blocks; a block is a list of natural numbers. `IsKQS v R` states conditions 1–3 of the definition literally: every block has length 4, has no duplicates and lies in $\{0, \dots, v-1\}$; for each class C and each $x < v$, the number of blocks of C containing x is 1; and for all $x \neq y$ below v , the number of blocks of R containing both is 1. `HasKQS v` is $\exists R, \text{IsKQS } v \text{ } R$.
- **Exceptional** v is $v \bmod 12 = 4 \wedge v < 316 \wedge \neg \text{HasKQS } v$. **ExactlyExceptional** k says that the exceptional values are exactly the entries of a duplicate-free list of length k . `Clause1`, `Clause2` (that is, `ExactlyExceptional 21`) and `Conjecture = Clause1  $\wedge$  Clause2` formalize the statement. `ExceptionalLe` and `ExactlyExceptionalLe` formalize the non-strict variant.
- `check v R` is a fast Boolean checker that uses bitmasks ($\sum_{a \in B} 2^a$). For each class, the block masks must be pairwise disjoint and cover $\{0, \dots, v-1\}$. For each point x , the masks of the blocks through x , with bit x cleared, must be pairwise disjoint and cover all $y \neq x$. The theorem `check_sound : check v R = true  $\rightarrow$  IsKQS v R` is proved in general, via the counting lemma `disjUnion_sound`: if a disjointness-checked union succeeds, then the number of masks having bit y , plus bit y of the start value, equals bit y of the result.
- `R4, \dots, R76` are the seven resolutions, written out as explicit lists. `kqs_4, \dots, kqs_76` prove `IsKQS v Rv` by `check_sound` and kernel evaluation (`decide +kernel`).
- `exceptional_le_19`: any duplicate-free list of exceptional values has length at most 19. `not_exactlyExceptional`: no $k \geq 20$ is the exact count. `clause2_false :  $\neg$ Clause2`, and `conjecture_00000001028_false :  $\neg$ Conjecture`. `exceptionalLe_le_20` and `clause2_le_false` handle the non-strict reading. `admissible_below_316` lists the 26 admissible values below 316.
- Non-vacuity: `not_hasKQS_two_three` shows that `HasKQS` is not trivially true (no KQS on 2 or 3 points), and `exceptional_iff` shows that for admissible $v < 316$, `Exceptional v` is equivalent to $\neg \text{HasKQS } v$.

`lake build` succeeds with no errors or warnings (about one minute of kernel evaluation). There is no `sorry`, no `native_decide`, and no added axiom; `#print axioms` shows at most `propext`, `Classical.choice` and `Quot.sound`.

What is not in Lean. The geometric descriptions of the designs (affine spaces, unital, packing, Lemma 3) explain where the designs come from. They are not needed for the proof, because each design is verified directly as an explicit object. The necessary condition of Remark 2 (used only for the “any residue” reading) and the Hanani–Ray–Chaudhuri–Wilson theorem are not formalized.

6 Independent check

`verify.py` (Python 3 standard library) reads the seven resolutions from `Main.lean`. It checks them by explicit pair counting, which is a different method from the Lean bitmask checker. It also re-derives each design from its description: lines of $\text{AG}(2, 4)$ and $\text{AG}(3, 4)$, secant lines of the

Hermitian curve over $\mathbb{F}_9$, lines of $\text{PG}(3, 3)$, and the developments of Lemma 3, including conditions (a) and (b). Finally, it recounts the admissible values and the bound.

7 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py           # Python 3 standard library
pdflatex report.tex
```

References

- [1] H. Hanani, D. K. Ray-Chaudhuri and R. M. Wilson, On resolvable designs, *Discrete Math.* 3 (1972), 343–357.
- [2] R. H. F. Denniston, Some packings of projective spaces, *Atti Accad. Naz. Lincei Rend. Cl. Sci. Fis. Mat. Natur.* 52 (1972), 36–40.
- [3] C. J. Colbourn and J. H. Dinitz (eds.), *Handbook of Combinatorial Designs*, 2nd ed., Chapman & Hall/CRC, 2007 (resolvable designs, unitals, 1-rotational constructions).